

Aryan Koul

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EDUCATION

The Pennsylvania State University, Pennsylvania, USA

May 2024

Majors: BSc. Computer Science

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Database Systems, **Honors:** Dean's List (2020)

WORK EXPERIENCE

AlumAi Inc | Developer | New York

Sep 2024 - till date

- Developed Python-based AI projects, including game-playing bots (e.g., Tic Tac Toe), focusing on clean logic, modular design, and maintainability
- Applied best practices for code readability and documentation, including consistent use of inline comments and modular design
- Built a nutrition meal planner application for android integrating the OpenAI API to generate personalized meal plans based on user inputs (age, fitness goals, preferences), featuring responsive UI, real-time nutrition info, and robust async API handling for smooth user experience.
- Developed a data analytics application for android supporting CSV/JSON ingestion, statistical analysis, and dynamic data visualizations (bar, pie, line charts) using chart libraries, optimized for large datasets with frontend-backend data binding and error validation.
- Developed a career counseling app integrating OpenAI for personalized university and career recommendations based on user input, with Firebase for authentication, profile management, and real-time mentor chat.

Engineering Capstone Project | Team Member | PSU

Jan 2024 - Apr 2024

- Collaborated with a team to design innovative runway de-icing solutions for airports
- Applied systems thinking and engineering principles to solve real-world logistical challenges
- Worked in a cross-functional team spanning software, economics, and engineering disciplines

Database Management System Project | Developer | PSU

Sep 2023 – Dec 2023

- Designed and implemented an inventory management system for a plant store
- Created backend architecture using MySQL for efficient data storage and retrieval
- Collaborated with a partner and incorporated peer feedback to align SQL queries with frontend requirements

Independent Projects | AI and Game Bots Development | PSU

Sep 2023 – Apr 2024

- Built AI-based bots for games such as Chess and Tic Tac Toe using Python
- Implemented decision-making algorithms like Dijkstra's and explored computational theory and ML concepts

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, MySQL

Tools & Libraries: Git, Jupyter Notebook, VS Code

Course Projects: AI Game Bots, Inventory Systems, De-icing Engineering Models

CERTIFICATIONS & WORKSHOPS

AI Workshops in Python

Apps info:

Nutrition Meal Planner App

Technical Skills:

- **API Integration & Management** - Integrated OpenAI API for intelligent meal plan generation and real-time nutritional analysis
- **User Input Validation & Processing** - Implemented comprehensive form handling for user demographics, dietary preferences, and fitness goals
- **Dynamic Content Generation** - Built interactive meal planning system with on-demand nutritional information retrieval
- **Frontend Development** - Created responsive user interfaces for data input, meal display, and nutrition information presentation
- **State Management** - Managed complex application state across multiple user interactions and API calls
- **Asynchronous Programming** - Handled multiple API requests efficiently with proper error handling and loading states
- **Data Parsing & Display** - Processed and formatted API responses for user-friendly meal plans and nutritional breakdowns
- **User Experience Design** - Designed intuitive workflows for personalized meal planning and nutrition tracking

Data Analytics App

Technical Skills:

- **File Processing & Data Ingestion** - Built robust CSV and JSON file upload and parsing functionality
- **Data Visualization** - Created interactive bar charts, pie charts, and line charts for comprehensive data representation
- **Statistical Analysis & Insights** - Implemented algorithms to identify key metrics like highest values, frequency analysis, and data trends
- **Data Transformation** - Processed raw data into meaningful visualizations and statistical summaries
- **Frontend Data Binding** - Connected data processing backend with dynamic chart rendering on the frontend
- **Performance Optimization** - Optimized data processing for large datasets with efficient parsing and rendering

Analytical Skills:

- **Business Intelligence** - Generated actionable insights from raw data, including trend analysis and key performance indicators
- **Data Interpretation** - Translated complex datasets into clear, understandable visual narratives
- **Automated Reporting** - Built systems to automatically identify and highlight significant data patterns and outliers.

Both apps demonstrate ability to build full-stack applications that integrate external APIs, process user data, and deliver intelligent, personalized experiences.